

Exp. 5. Band Structure Simulation for Graphene and CNTs using NanoHub

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Slot-c ECA3102 NANO ELECTRONICS

OUTPUT

Assembly of Basic Applications for Coordinated Understanding of Semiconductors

Choose Applications and Tools in this Dropdown Menu or from Images

Bandstructure Lab - Bulk, quantum wells, nanowires

Homework: Bandstructure Lab x Band Structure Lab

1 Device Type → 2 Physics → 3 K-Space → 4 Numerics → 5 Simulate

Geometry: Nanowire (1D-periodic)

Calculation For: Electrons

Material: Si

Alloy Model: Virtual Crystal Approximation

Mass Fraction: 0.5

Job Type: Calculate the wire band structure

Device cross section: Circle

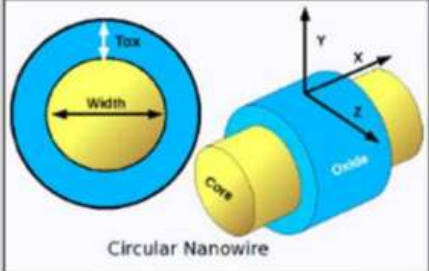
Diameter: 2.1nm

oxide Thickness(Tox): 2.2nm

Crystal Orientation

Transport direction (X): (100)

Confinement direction (Z): [010]



Circular Nanowire

Physics >

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Bandstructure Lab - Bulk, quantum wells, nanowires

Homework Bandstructure Lab x

Band Structure Lab

1 Device Type

2 Physics

3 K-Space

4 Numerics

5 Simulate

Electronic Structure

Strain

Tight Binding Model: sp3d5s*

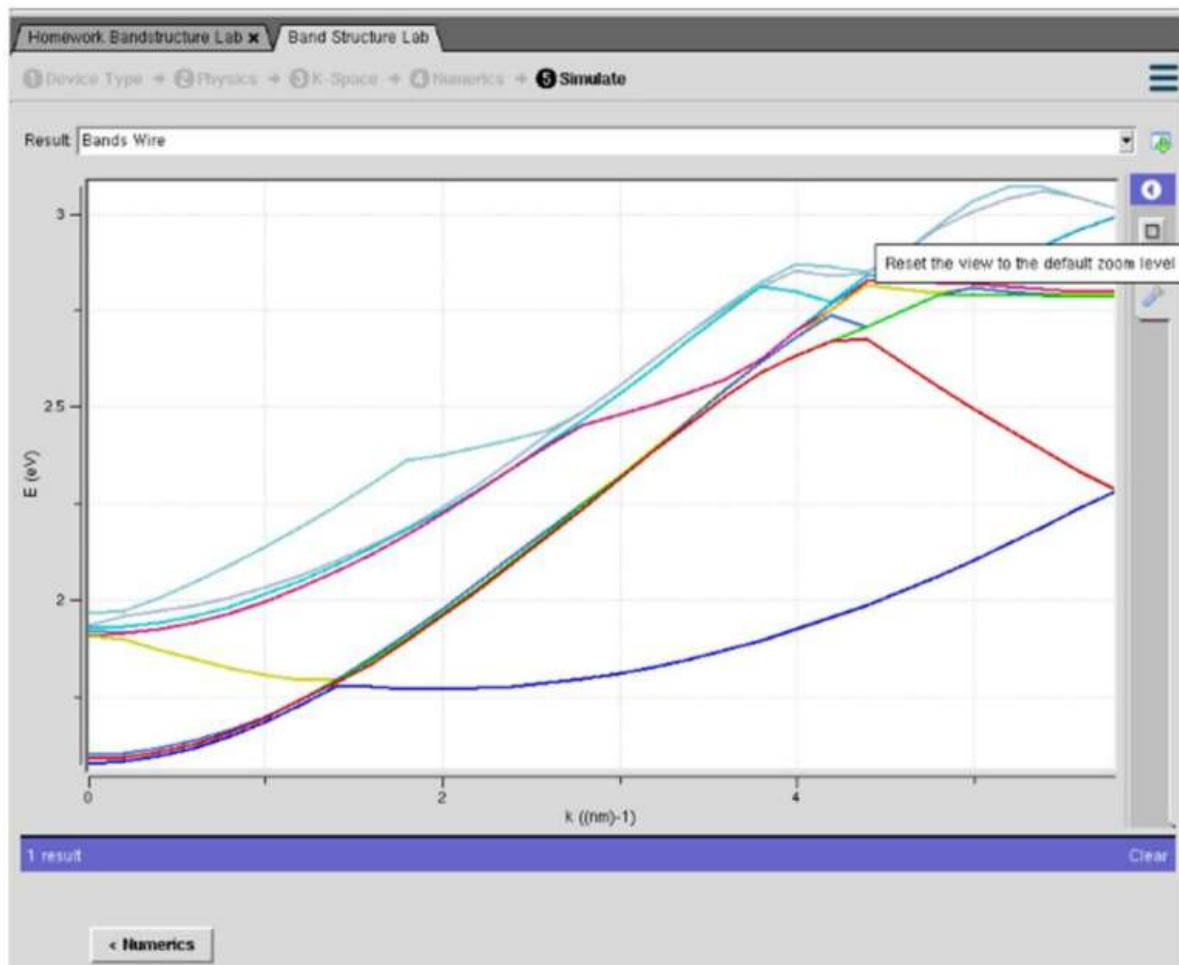
Spin-Orbit Coupling: ☒ ☐ no

Dangling Bond Energy: 30eV

Explore: Conduction bands

< Device Type

K-Space >



These figures show the simulation of a silicon nanowire band structure. The first image configures the nanowire geometry and material properties, the second sets the electronic structure parameters, and the third displays the band structure (Energy vs. Wave Vector), illustrating the allowed electron energy bands in the nanowire.

Case Study 1: Simulate the band-structure of InAs circular nanowire

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Homework Bandstructure Lab x Band Structure Lab

1 Device Type + 2 Physics + 3 K-Space + 4 Nanorcs + 5 Simulate

Geometry: Nanowire (1D-periodic)

Calculation For: Electrons

Material: InAs

Atom Model: Virtual Crystal Approximation

Molar Fraction: 0.5

Job Type: Calculate the wire band structure

Device cross section: Circle

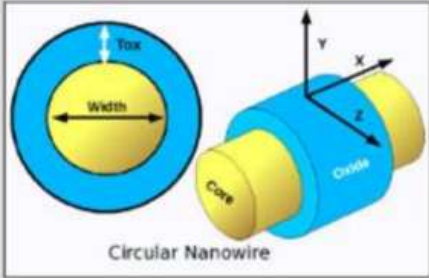
Diameter: 2.1nm

Core Thickness(Tox): 2.2nm

Crystal Orientation

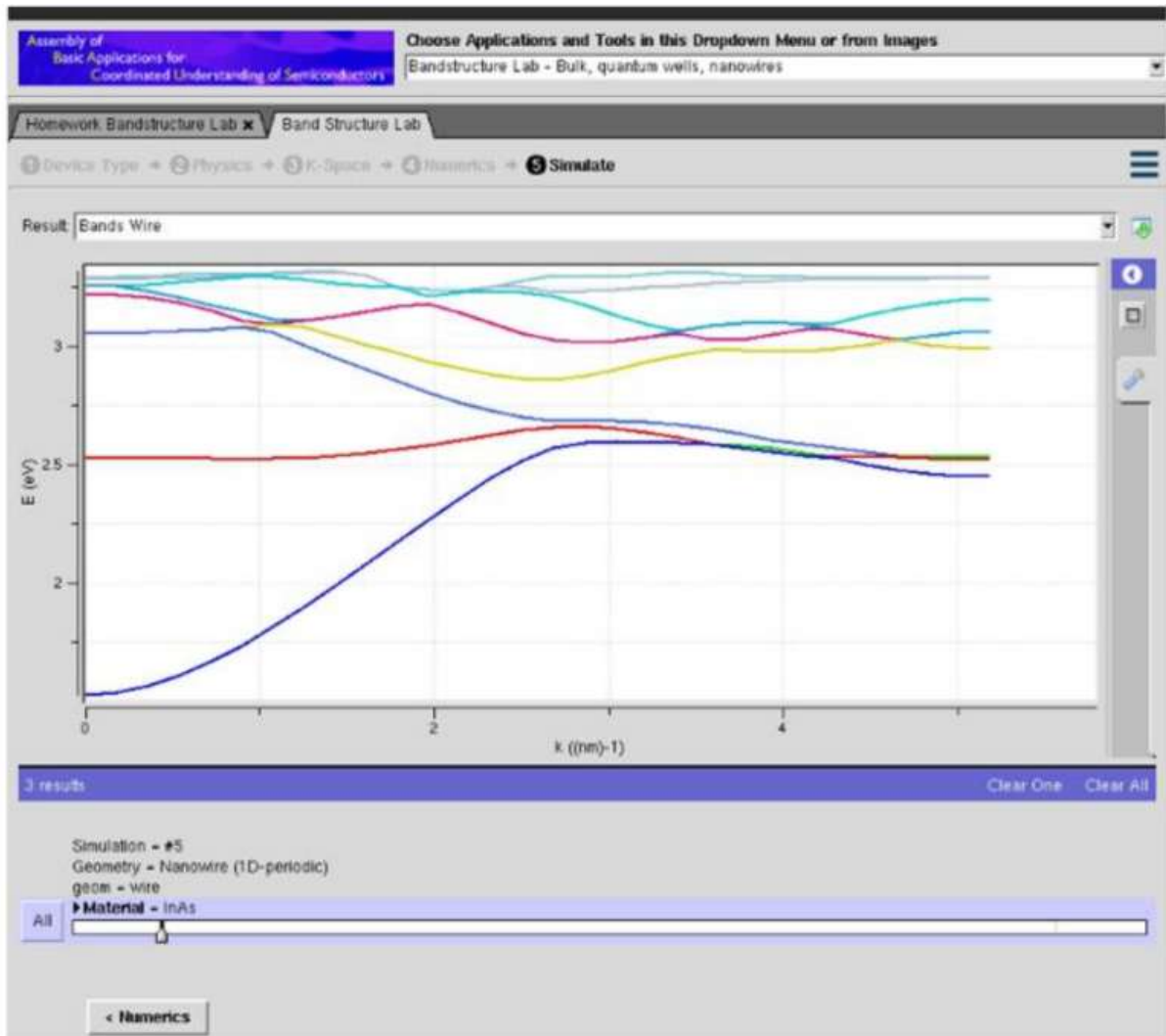
Transport direction (X): [100]

Confinement direction (Z): [010]

The diagram illustrates the geometry of a circular nanowire. On the left, a circular cross-section is shown with a central yellow circle labeled 'Core' and an outer blue ring labeled 'Oxide'. The width of the core is indicated by a double-headed arrow labeled 'Width', and the thickness of the oxide layer is indicated by a double-headed arrow labeled 'Tox'. On the right, a 3D perspective view of the nanowire is shown, with a coordinate system (X, Y, Z) indicating the transport direction (X) and confinement direction (Z). The core is yellow and the oxide is blue.

Circular Nanowire

Physics >



These images show the setup and simulation result of a nanowire band structure using semiconductor simulation software. The graph displays the energy bands of electrons, which helps analyze the nanowire's electrical properties.

Case Study 2: Simulate the band-structure of AlGaAs circular nanowire

Assembly of
Basic Applications for
Coordinated Understanding of Semiconductors

Choose Applications and Tools in this Dropdown Menu or from Images
Bandstructure Lab - Bulk, quantum wells, nanowires

Homework: Bandstructure Lab x Band Structure Lab

1 Device Type → 2 Physics → 3 K-Space → 4 Numerics → 5 Simulate

Geometry: Nanowire (1D-periodic)

Calculation For: Electrons

Material: AlGaAs

Alloy Model: Virtual Crystal Approximation

Molar fraction: 0.5

Job Type: Calculate the wire band structure

Device cross section: Circle

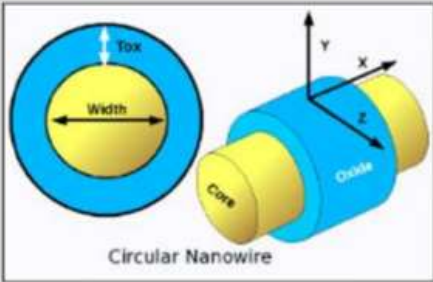
Diameter: 2.1nm

oxide Thickness(Tox): 2.2nm

Crystal Orientation

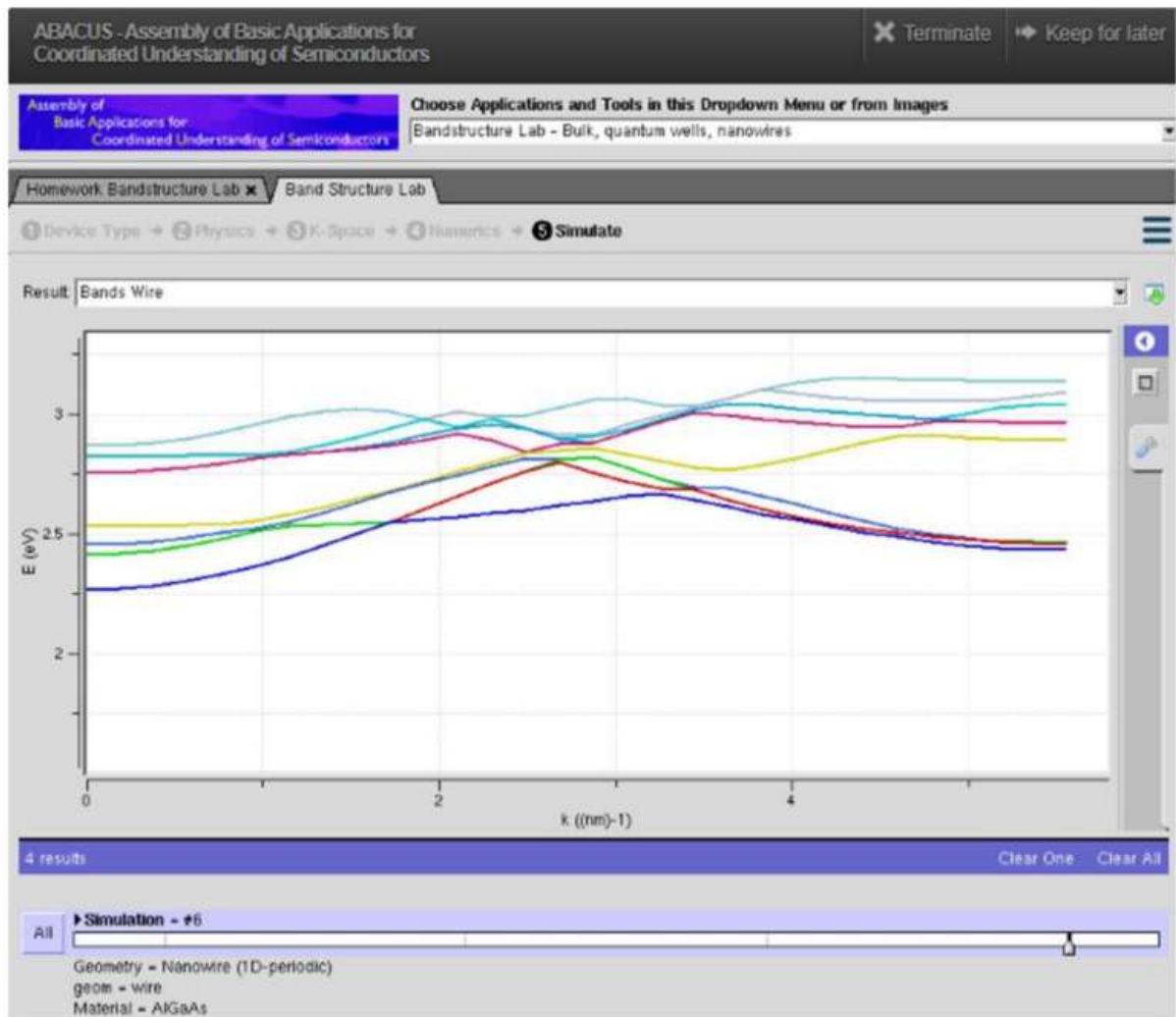
Transport direction (X): [100]

Confinement direction (Z): [010]



The diagram illustrates the geometry of a circular nanowire. On the left, a cross-sectional view shows a central yellow circle labeled 'Width' and an outer blue ring labeled 'Tox'. On the right, a 3D perspective view shows a yellow cylinder labeled 'Core' and a blue cylinder labeled 'Oxide' surrounding it. A coordinate system with x, y, and z axes is shown next to the 3D view. The text 'Circular Nanowire' is centered below the diagrams.

Physics >



These images show the setup and output of an AlGaAs nanowire band structure simulation.

The graph shows the electron energy bands, which are used to study the electrical properties of the nanowire.